QIUZHEN QUALIFY EXAM FOR ARTIFICIAL INTELLIGENCE

MOCK EXAMINATION NO. 2

Instructions

- This examination consists of four sections, each worth 33 points. Choose exactly three sections and answer all questions in the selected sections.
- Write your name and student ID on the answer sheet for 1 additional point. The maximum possible score is 100.
- State the three selected sections at the beginning of the answer sheet. If more than three sections are attempted, only the first three will be graded.
- Unless stated otherwise, justify every claim and show the essential derivation. Unsupported final formulas receive limited credit.

A. Machine Learning Theory

1. [3 pts.] For each item, select exactly one answer. Each item is worth 0.5 points.

(i) A rank-deficient least-squares problem can have multiple coefficient minimizers but a unique fitted vector $X\hat{w}$. (True / False)

(ii) Pruning a decision tree must reduce its training error. (True / False)

(iii) The conditions $k_m \to \infty$ and $k_m/m \to 0$ occur in classical consistency results for k-NN. (True / False)

(iv) Every underdetermined linear system has a unique sparsest solution. (True / False)

(v) If a matrix satisfies the RIP of order $2s$ with constant below 1, two distinct s-sparse vectors cannot have identical measurements. (True / False)

(vi) Data-dependent feature standardization may be fitted on the full train-test collection without causing leakage. (True / False)

2. [8 pts.] Let $X \in \mathbb{R}^{m \times d}$ have singular value decomposition $X = U\Sigma V^\top$. Consider

$$J_\lambda(w) = \frac{1}{2m}\|Xw - y\|_2^2 + \frac{\lambda}{2}\|w\|_2^2, \qquad \lambda > 0.$$

Derive the unique minimizer in both matrix form and singular-vector coordinates. Identify the shrinkage factor applied to each right-singular direction, explain what happens in $\ker(X)$, and determine the limit as $\lambda \downarrow 0$.

3. [7 pts.] A large decision tree is trained on one sample. From it, a deterministic pruning procedure produces a finite collection $\mathcal{P}$ of M candidate subtrees. An independent validation sample of size n is used to select $\hat{T} \in \arg\min_{T \in \mathcal{P}} \widehat{L}_{\text{val}}(T)$ under zero-one loss. Derive a high-probability bound comparing $L(\hat{T})$ with $\min_{T \in \mathcal{P}} L(T)$. State precisely where independence is used and explain why the same argument is invalid if the validation labels influence construction of $\mathcal{P}$.

4. [7 pts.] Let $X, X' \in \mathbb{R}^d$ be independent random vectors and define $D^2 = \|X - X'\|_2^2 = \sum_{j=1}^d Z_j$. Assume the Z_j are independent and identically distributed with mean $\mu > 0$ and variance $\nu < \infty$.

(a) **[3 pts.]** Compute $\mathbb{E}[D^2]$, $\mathrm{Var}(D^2)$, and the coefficient of variation of D^2.

(b) **[2 pts.]** Explain the implication for nearest-neighbor discrimination in high dimension without claiming an unconditional theorem for every distribution.

(c) **[2 pts.]** State classical asymptotic conditions on $k = k_m$ used for k-NN consistency, and state how feature preprocessing must be fitted in an evaluation pipeline.

5. [8 pts.] For $A \in \mathbb{R}^{q \times d}$, define the null-space property (NSP) of order s by

$$\|h_T\|_1 < \|h_{T^c}\|_1 \quad \text{for every } h \in \ker(A) \setminus \{0\} \text{ and every } T \subseteq [d],\ |T| \le s.$$

Prove that A has the NSP of order s if and only if every s-sparse vector x is the unique solution of

$$\min_z \|z\|_1 \quad \text{s.t.} \quad Az = Ax.$$

B. Deep Learning and Reinforcement Learning

1. [9 pts.] An image $X \in \mathbb{R}^{64 \times 64 \times 3}$ is processed by two convolutional layers. Layer 1 has 16 output channels, kernel 3×3, stride 2, dilation 1, and padding 1. Layer 2 has 32 output channels, kernel 3×3, stride 1, dilation 2, and padding 2. Both layers use one bias per output channel.

(a) **[3 pts.]** Compute both output shapes and the total number of trainable parameters.

(b) **[3 pts.]** Compute the receptive-field width and effective jump, measured in input pixels, after each layer.

(c) **[3 pts.]** State the conditions under which convolution is translation equivariant. Explain separately how stride 2 and zero-padding boundaries affect exact equivariance.

2. [8 pts.] Let unit embeddings be used in a supervised contrastive loss. For anchor z, let P be its nonempty positive index set and A the set of all other examples in the batch. With temperature $\tau > 0$,

$$\ell(z) = -\frac{1}{|P|} \sum_{p \in P} \log \frac{\exp(z^\top z_p/\tau)}{\sum_{a \in A} \exp(z^\top z_a/\tau)}.$$

Treat the other embeddings as fixed and derive $\nabla_z \ell$. Interpret the result, explain the role of τ, and describe one principled response to false negatives. State how the unit-norm constraint changes the geometric interpretation of the raw gradient.

3. [9 pts.] For the original GAN objective

$$\min_G \max_D \ \mathbb{E}_{x \sim p_{\text{data}}} \log D(x) + \mathbb{E}_{x \sim p_g} \log(1 - D(x)),$$

assume p_{data} and p_g have densities with respect to a common measure.

(a) **[3 pts.]** Derive the optimal discriminator for fixed G.

(b) **[3 pts.]** Substitute it into the objective and express the result using the Jensen–Shannon divergence.

(c) **[3 pts.]** Compare the generator gradients of the saturating loss $\log(1 - D(G(z)))$ and the non-saturating loss $-\log D(G(z))$ when the discriminator rejects generated samples with high confidence. State why the idealized divergence result does not prove convergence of alternating SGD.

4. [7 pts.] For a finite discounted MDP, write the tabular SARSA and Q-learning updates. Show which Bellman operator each one targets in conditional expectation, classify each method as on-policy or off-policy, and state sufficient exploration and step-size conditions for the classical tabular Q-learning convergence result. Explain why that result does not directly establish convergence of a nonlinear deep Q-network.

C. Optimization Methods in Artificial Intelligence

1. [10 pts.] Let $X \subseteq \mathbb{R}^d$ be nonempty, closed, and convex, and let f be convex and L-smooth. Projected gradient descent is

$$x^+ = \Pi_X(x - \gamma \nabla f(x)).$$

(a) **[3 pts.]** Derive the variational inequality that characterizes the projection.

(b) **[3 pts.]** Prove that $x^\star$ minimizes f over X if and only if $x^\star = \Pi_X(x^\star - \gamma \nabla f(x^\star))$ for any $\gamma > 0$.

(c) **[4 pts.]** For $0 < \gamma < 2/L$, prove a quantitative descent inequality in terms of the gradient mapping $G_\gamma(x) = (x - x^+)/\gamma$.

2. [10 pts.] Let F be convex on a closed convex set X. Assume $\|g\| \le G$ for all $g \in \partial F(x)$ and $\left\|x^0 - x^\star\right\| \le R$. For

$$x^{k+1} = \Pi_X(x^k - \eta g^k), \qquad g^k \in \partial F(x^k),$$

derive a bound on $F(\bar{x}^K) - F(x^\star)$ for $\bar{x}^K = K^{-1} \sum_{k=0}^{K-1} x^k$. Choose a constant step size to obtain an explicit $O(1/\sqrt{K})$ rate. State why the proof does not imply the same rate for the last iterate.

3. [13 pts.] Let $f(x) = \frac{1}{2} x^\top Q x$ with $Q \succ 0$ and eigenvalues in $[\mu, L]$. Heavy-Ball uses

$$x^{k+1} = x^k - \gamma \nabla f(x^k) + \beta(x^k - x^{k-1}), \qquad 0 \le \beta < 1.$$

(a) **[4 pts.]** Diagonalize the recurrence and derive the scalar characteristic polynomial for an eigenvalue λ.

(b) **[5 pts.]** Prove that both roots lie strictly inside the unit disk for every $\lambda \in [\mu, L]$ if and only if $0 < \gamma < 2(1 + \beta)/L$.

(c) **[4 pts.]** For $Q = \mathrm{diag}(1, 25)$, $\beta = 0.6$, and $\gamma = 0.1$, determine stability and compute the modulus of the roots in both eigendirections.

D. Natural Language Processing

1. [6 pts.] A trigram model has vocabulary size 4. For the history (a, b), the count is $C(a, b) = 4$ and $C(a, b, c) = 2$. Also $p_{\text{bi}}(c \mid b) = 0.4$ and $p_{\text{uni}}(c) = 0.25$.

(a) **[2 pts.]** Compute the trigram MLE and add-one estimate of $p(c \mid a, b)$.

(b) **[2 pts.]** Compute the interpolated probability using weights $(0.5, 0.3, 0.2)$ on trigram, bigram, and unigram probabilities.

(c) **[2 pts.]** A two-token test continuation is assigned probabilities 0.42 and 0.35. Compute its per-token perplexity and state why perplexities from different tokenizations are not directly comparable.

2. [9 pts.] Consider a first-order HMM with K hidden labels and an observation sequence of length T.

(a) **[3 pts.]** Write the factorized joint distribution and give the supervised maximum-likelihood estimates of transition and emission probabilities from labeled sequences.

(b) **[3 pts.]** Derive the forward recursion for the observation likelihood.

(c) **[3 pts.]** Derive the Viterbi recursion and backtracking rule for MAP label decoding. Give time and memory complexities.

3. [8 pts.] Suppose an empirical scaling law is

$$L(P,T) = L_\infty + AP^{-\alpha} + BT^{-\beta}, \qquad A, B, \alpha, \beta > 0,$$

and the dominant training-compute budget imposes $PT = C$. Derive the optimal allocations $P^*(C)$ and $T^*(C)$, including their constants, and derive the resulting power-law exponent of the excess loss. State two assumptions or caveats needed before using this calculation to predict a real training run.

4. [10 pts.] A knowledge base contains typed entities, relation triples, and textual descriptions. It is incomplete, and the system must recommend missing tail entities for a query $(h, r, ?)$. Design a system that combines a TransE-style graph scorer with a text encoder. Specify training examples and type-aware negative sampling, objectives and parameter constraints, candidate generation and reranking, calibration or abstention, dominant complexity terms, and a filtered link-prediction evaluation protocol. Include failure analyses for false negatives and one-to-many relations.